UNIT 2 · 9 hours

Types and Construction of Single-Phase Transformers



Transformers and Electrical Machines

Basic Electrical and Electronics Engineering (EEE103)
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Learning objectives

- Name the main parts of a transformer
- Distinguish core type from shell type
- Explain the purpose of laminations and silicon steel
- State common classifications of transformers

Syllabus context

Transformers: necessity of transformers, principle of operation, types and construction of single-phase transformers, EMF equation, losses and efficiency.

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Main parts

Core: stack of silicon-steel laminations (0.35–0.7 mm), each insulated, forms the magnetic path; limbs and yokes.

Windings: primary connected to supply, secondary to load; copper or aluminium, insulated.

Insulation: paper, varnish, press-board between turns and windings.

Tank and oil (large units): cooling and insulation. Bushings bring out terminals.

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Why laminate the core and use silicon steel?

Eddy-current loss

Changing flux induces currents in the iron. Thin insulated laminations break their paths, reducing I^2R loss.

Hysteresis loss

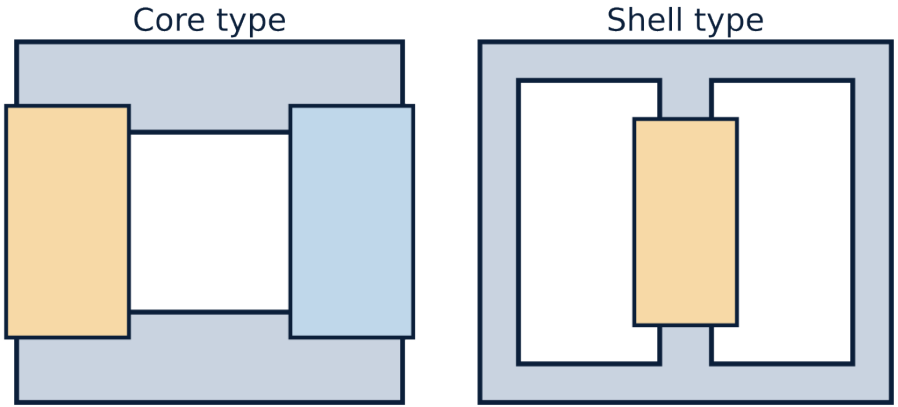
Silicon steel has a narrow hysteresis loop (about 3–4 % Si), so less energy is lost per cycle.

High permeability

Provides a low-reluctance path so little magnetising current is needed.

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Core type and shell type



- Core type: windings surround the limbs; one magnetic path; half of each winding on each limb to cut leakage.
- Shell type: windings on the central limb; core surrounds them; two parallel flux paths.

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Core type versus shell type

Feature	Core type	Shell type
Magnetic circuit	Single path	Two parallel paths
Windings	On two limbs	On central limb
Leakage flux	Higher	Lower
Winding cooling / repair	Easy (exposed)	Harder (enclosed)
Typical use	High-voltage power transformers	Low-voltage, small and electronic

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Classification of transformers

- By function: step-up and step-down.
- By use: power, distribution, instrument (CT and PT), isolation, auto-transformer.
- By cooling: dry (air) type and oil-immersed.
- By phase: single-phase and three-phase.

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Key takeaways

- Core of insulated silicon-steel laminations reduces eddy and hysteresis loss
- Core type: coils around limbs; shell type: core around coils
- Primary is connected to the supply, secondary to the load
- Large transformers are oil-cooled

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Practice questions

Q1. Why is silicon steel used for the transformer core?

Q2. Compare core type and shell type transformers.

Q3. What is the thickness of a typical lamination and why?

Q4. Name three types of transformers by application.